Dual-Channel Pete Millett Interface Replica

Universal 150 V_{RMS} Active Differential Probe & Audio-Band Conditioning AFE

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Document Control & Baseline Specifications

Target Hosts:	M-Audio Solo, MOTU M4, SDS804X HD	**Document ID:**	EDD-PMSI-UNIV-003
Topology:	Dual-Channel Galvanically Isolated AFE	**Revision:**	3.1 (Consolidated Freeze)
Date:	October 2026	**Classification:**	Open Engineering Baseline
Status:	Pre-Fabrication Release	**Primary Bandwidth:**	3.35 Hz – 2.5 MHz

Table 1: Document control and target host integration baseline.

System Parameter	Hardware Implementation Target
Architecture Format	Dual-Channel (Stereo Ch 1 / Ch 2), fully mirrored analog signal paths
Coupling Mode	Strictly AC-Coupled via 0.47 μF 400 V Metallized Polypropylene Film ($f_{-3\text{dB}} \approx 3.35$ Hz, multi-pitch footprint with 8.5 mm width clearance)
Maximum Continuous Voltage	150 V_{RMS} continuous (212 V_{peak}) across all attenuator ranges (100% duty cycle)
Maximum Intermittent Voltage	250 V_{RMS} intermittent (353 V_{peak}) for up to 5 minutes duration
Wideband Scope Tap Bandwidth	Flat to within −0.5 dB at 2.0 MHz on 1× and 10× ranges; −3 dB corner ≥ 2.5 MHz
Attenuator Ladder Network	101.0 kΩ total (R_{302} = 90.9 kΩ, R_{303} = 9.09 kΩ, R_{304} = 909 Ω in 0.6 W 500 V Vishay CMF55 axial film mounted on switch lugs; R_{305} = 102 Ω on PCB)
Attenuator Range Switch	Front-panel SP4T Break-Before-Make (BBM) rotary switch with solder lugs (E-Switch KC14A13.001NLS or C&K A10405RNZQ, 1,000 V_{RMS} dielectric withstand)
Switch-to-PCB Interconnect	3-wire PTFE unshielded harness (Guarded Trio) to a 5-pin right-angle PCB header (Pins 1, 3, 5 active; Pins 2, 4 grounded on PCB and de-pinned)
Overvoltage Current Limiter	Socketed Vishay NFR25 4.7 kΩ 0.5 W flameproof fusible resistor (R_{312}, R_{313}) on 15.24 mm pitch with vertical 1.5 × 6.0 mm PCB air-gap isolation slot
Input Rail Steering Clamps	Nexperia BAV199 SOT-23 ultra-low-leakage dual series diodes (D_{309}, D_{305}, $I_R \leq 5$ pA, $C_d \leq 1.5$ pF) steering fault current to ±15 V rails
Rail Overvoltage Protection	SMAJ16A unidirectional TVS diodes (D_{105}, D_{106}) clamping the ±15 V rails against rail pumping
Active Input Buffers	Dual OPA1656 CMOS Op-Amps (U_{302}, 10 pA bias, 2.9 nV/$\sqrt{\text{Hz}}$, 53 MHz) with 4.7 pF C0G feedback caps (C_{308}, C_{309}) + spare parallel DNP pads
Difference Amplifier Stage	Texas Instruments INA134 (U_{303}, Pin 1 tied directly to AGND, > 90 dB guaranteed CMRR)
Selectable Output Clamping	Off-signal BAV199 steering bridge (D_{306}) with 3-pin jumpers (JP_{301}, JP_{302}) selecting BAV199 (D_{301}, ±1.4 V_{pk} Solo) or 1SMB5917 4.7 V Zeners (±5.4 V_{pk} M4/Scope)
Universal Output Stages	Simultaneous Balanced TRS (DRV135, $G = 2$) and Unbalanced BNC (J_{303}, $G = 1$ via 200 Ω R_{311})
Input Ground Switching	Permanently eliminated; full floating differential input prevents Earth short-circuits
DC Galvanic Isolation	XP Power JCA0305D03 (3 W, 5 V in, ±15 V out, 1,500 VDC isolation, Faraday shield)
Primary DC Power Crowbar	USB-C (6-pin power) with 1812 SMD PTC fuse (1.1 A) + SMCJ6.0A TVS clamp
Power Switching & Indicators	Dual-pole 5V Master switch (SW_{102}) + Dual-pole ±15V Main switch (SW_{101}) with 3-LED status display (D_{102}, D_{103}, D_{104})
Chassis Ground Lug Network	Isolated analog AGND referenced to enclosure (J_{102}) via parallel 1 MΩ 1 W bleed resistor (R_{104}) + 10 nF 1 kV RF capacitor (C_{107})

Table 2: Consolidated system electrical, mechanical, and safety specifications.

1 Executive Design Summary & Tri-Host Architecture

This document establishes the electrical, mechanical, and PCB layout specifications for an optimized dual-channel replica of the classic Pete Millett Soundcard Interface. While originally conceived as a single-channel, through-hole audio interface accessory for consumer PC soundcards, this Revision 3.1 architecture transforms the design into a **high-precision, dual-channel, galvanically isolated active differential probe and wideband signal conditioning front-end**.

The system is engineered to serve three distinct instrumentation hosts interchangeably without hardware reconfiguration:

1. **M-Audio Solo (Consumer 16-Bit USB Audio Interface):** Provides basic single-ended and balanced audio-band spectrum analysis, serving as an isolated, sacrificial signal source and digitizer (20 Hz – 20 kHz, ≈ 92–94 dB dynamic range). Because the Solo line input clips near +2 dBu (≈ 1.0 V_{RMS}), the interface provides a dedicated Low-Clamp jumper position (±1.4 V_{pk} at the single-ended 'SCPROT' node).

2. **MOTU M4 (Professional 24-Bit / 192 kHz Interface):** Connects via dedicated rear-panel balanced Line Inputs 3 & 4 (+18 dBu headroom, fixed analog gain) to perform high-resolution automated harmonic distortion (THD+N down to −110 dB), intermodulation, and phase tracking using software suites such as ARTA (STEPS, LIMP), REW, and Multitone.

3. **Siglent SDS804X HD (12-Bit High-Resolution Bench Oscilloscope):** Functions as a $1,500$ **VDC isolated, high-voltage active differential probe**. In High-Resolution (ERES) mode, the oscilloscope achieves 14 to 16 effective bits (ENOB) with continuous bandwidth from 3.35 Hz past 2.0 MHz, making it ideal for inspecting ultrasonic oscillation, power-supply rail ripple, slew-rate transients, and Class-D amplifier switching noise without ground-loop hazards.

1.1 Key Design Modifications from Baseline

- **Dual-Channel Topology:** Complete mirrored duplication of attenuators, fusible limiters, OPA1656 buffers, INA134 difference stages, clamping bridges, and DRV135 drivers for stereo operation.
- **Power Switching & Kick-Back Protection:** Added a primary Master Switch (SW_{102}) and secondary Dual-Rail Switch (SW_{101}), a 3-LED status monitoring network, and dual secondary SMAJ16A TVS clamping diodes (D_{105}, D_{106}) across the ±15 V rails to absorb kickback current during high-voltage input faults.
- **Chassis Ground Isolation Lug:** Integrated a dedicated chassis bonding network comprising a 1 MΩ 1 W bleeder resistor in parallel with a 10 nF 1 kV ceramic capacitor connecting chassis earth to isolated analog ground for RF shielding and electrostatic drainage without ground loops.
- **Transition to Vishay NFR25 Fusible Resistors:** Replaced both Pete's original noisy 50 kΩ resistors and intermediate depletion FET concepts with socketed 4.7 kΩ flameproof fusible resistors (R_{312}, R_{313}), lowering thermal noise to 8.82 nV/$\sqrt{\text{Hz}}$ while ensuring safe open-circuit disconnection under catastrophic fault conditions.
- **BAV199 Low-Leakage Diodes:** Replaced legacy 1N4148 diodes with BAV199 dual series pairs ($I_R \leq$ 5 pA, $C_d \leq 1.5$ pF) backed by SMAJ16A TVS power-rail clamps.

- **Switch-Mounted 0.6 W Attenuator Ladder:** Re-engineered the precision divider ladder (90.9 kΩ, 9.09 kΩ, 909 Ω) to solder directly across the rotary switch solder lugs in a radial halo, completely eliminating high-frequency wire-to-wire crosstalk.
- **3-Wire PTFE Guarded Interconnect & 5-Pin Header:** Replaced multi-conductor ribbon cables and lossy coaxial cables with 3 unshielded PTFE wires routed to an interleaved 5-pin header, providing > 5.08 mm high-voltage creepage and grounded electrostatic shielding.
- **Buffer Loop Optimization (2 MHz):** Sized OPA1656 feedback capacitors to 4.7 pF C0G/NP0 (C_{308}, C_{309}) with spare parallel DNP pads, pushing the loop transmission pole to 4.51 MHz.
- **Elimination of Input Ground Switch (SW5) & Monitor Switch (SW3):** Hardwired as a permanently floating differential front-end to eliminate the risk of accidental short-circuits to Mains Earth Ground when paired with a bench oscilloscope.
- **Elimination of DC Zero-Adjust Potentiometer (R_{25}):** Tied INA134 Pin 1 directly to local isolated analog ground (AGND), preserving > 90 dB CMRR.
- **Dual-Level Off-Signal Zener Clamping Bridge:** Relocated voltage-defining Zeners into DC bias legs selecting between ±1.4 V_{pk} (Solo protection) and ±5.4 V_{pk} (M4 line level / Scope dynamic range) without placing non-linear Zener capacitance across the audio signal path.
- **Universal Output Connectivity:** Simultaneous balanced 1/4″ TRS jacks (via DRV135, $G = 2$) and unbalanced BNC jacks (via 200 Ω isolation resistor R_{311}).

2 System Architecture & Cross-Host Metrology

Table 3 outlines the operating modes, converter architectures, and metrology boundaries across the supported bench hosts.

Metric	M-Audio Solo	MOTU M4 (Ch 3/4)	Siglent SDS804X HD
Primary Role	Portable / Sacrificial Audio In	Reference Audio Distortion Rig	High-Voltage Floating Active Probe
Converter Type	16-bit ΔΣ Audio ADC	24-bit ESS Sabre32 ADC	Native 12-bit Pipeline ADC
Maximum Linear Input	+2 dBu (≈ 1.0 V_{RMS})	+18 dBu (6.15 V_{RMS})	Full screen ±40 V (at 5 V/div)
Hardware Dynamic Range	92 – 94 dB	**115** dB (ENOB ≈ 18.8)	65 – 70 dB (up to 15b in ERES)
Usable Bandwidth	10 Hz – 21.5 kHz	10 Hz – 90 kHz	**3.35** Hz – **2.5** MHz
Broadband Noise Floor	≈ 25 μV_{RMS}	≈ **10** μV_{RMS}	≈ 150 μV_{RMS} (2 MHz BW)
FFT Noise Floor (w/ Gain)	≈ −110 dBFS	**−145** to **−150** dBFS	≈ −95 dBm (Math FFT)
Interface Input Connection	Front 1/4″ Line (Bal/Unbal)	Rear 1/4″ TRS Line 3/4 (Bal)	Front BNC (1 MΩ ∥ 15 pF)
Ground Reference	Floating (USB laptop isolated)	Floating (USB bus powered)	**Mains Earth Grounded** (Chassis)
Interface Output Path	Balanced TRS (DRV135)	Balanced TRS (DRV135)	Rear BNC Tap (R_{311} = 200 Ω)
Clamp Selector Setting	Low (±1.4 V_{pk})	High (±5.4 V_{pk})	High (±5.4 V_{pk})

Table 3: Hardware capability and routing matrix across target instrumentation hosts.

2.1 Why the Oscilloscope Cannot Fully Replace the Soundcard

While the SDS804X HD in High-Res mode achieves an effective resolution of ≈ 15 bits with wide bandwidth, it cannot replace a dedicated audio interface for precision distortion testing:

- **Delta-Sigma Noise Shaping:** Audio converters (such as the ESS Sabre32 in the M4) oversample and push quantization noise outside the audio band, yielding an ultra-quiet floor down to -115 to -130 dBV.
- **Wideband Johnson Noise:** The oscilloscope front-end must maintain megahertz flat bandwidth. Even when band-limited, wideband thermal noise keeps the scope's noise floor near -88 dB. Op-amp or amplifier harmonics sitting at 0.0003% (-110 dB) disappear entirely beneath the scope's quantization floor.
- **Software Sweep Automation:** ARTA (STEPS) and REW automate multi-point stepped-sine distortion vs. frequency and power sweeps, automatically tracking harmonics HD2 through HD10. Oscilloscopes require manual data export and post-processing for similar audio metrics.

2.2 Why the Oscilloscope Requires This Active Front-End

Conversely, connecting an oscilloscope directly to an audio circuit via a standard 10X passive probe presents severe limitations:

a. **The Mains Earth Short Hazard:** The outer shield of the scope's BNC jack is hardwired to Mains Earth. Clipping a probe ground lead to a floating tube cathode resistor, a bridge-tied load (BTL) amplifier output, or an inverter rail creates a dead short to earth ground, instantly destroying circuit traces or the scope probe. The interface's 1,500 VDC galvanic isolation completely eliminates this hazard.

b. **High DC Bias Dynamic Range Loss:** Probing a 300 VDC vacuum tube plate carrying a 100 mVAC signal forces a standard scope to a 50 V/div or 100 V/div scale. The 100 mV audio ripple occupies a fraction of a single ADC quantization level. The interface's 400 V polypropylene film capacitors block the DC component completely, allowing the scope to operate on its most sensitive 20 mV/div scale.

c. **Capacitive Loading on High-Z Nodes:** A 10X passive probe has ≈ 15 pF of tip capacitance, which at 20 kHz represents an impedance of only 530 kΩ. On a high-impedance tube plate (100 kΩ), this causes high-frequency roll-off and phase shift. The interface's OPA1656 CMOS inputs buffer the node with negligible capacitance and 10^{13} Ω input resistance.

2.3 Comparison with DC Telemetry Front-End (TI INA228)

A dedicated DC telemetry circuit based on the Texas Instruments INA228 serves a fundamentally different measurement role than this wideband AC-coupled analog front-end:

- **Signal Domain:** The INA228 is an integrated 20-bit $\Delta\Sigma$ digital power/energy monitor strictly dedicated to DC bus voltage, shunt current, and slow thermal telemetry. This interface is an ultra-low-distortion, dynamic AC front-end with analog bandwidth extending to 2.5 MHz.

- **Common-Mode Ceiling:** The INA228 is restricted to an absolute maximum common-mode bus voltage of +85 VDC. This AFE withstands continuous 150 V_{RMS} (212 V_{pk}) and intermittent 250 V_{RMS} (353 V_{pk}), with a dielectric isolation barrier of 1,500 VDC between device-under-test and the host chassis.
- **Throughput & Interface:** The INA228 digitizes internally and outputs over an I^2C digital bus at conversion rates up to 50 kSPS. This interface provides real-time, zero-latency continuous analog waveforms to audio analyzers and high-speed bench oscilloscopes.

3 DC Power Supply, Rail Clamping & Enclosure Grounding

The power delivery network converts an unregulated +5 VDC supply into low-noise, fully isolated ±15 VDC analog split rails. The schematic architecture integrates dual-stage power switching, multi-LED status reporting, TVS rail protection against clamping kickback, and an AC-coupled chassis ground lug.

3.1 Power Input Connector & Dual-Pole Master Switch (SW_{102})

Power enters via a dedicated 6-pin USB-C receptacle (Power Only, J101):

- **Connector Pinout & CC Logic:** Provides two V_{BUS} power pins, two Ground pins, and dual Configuration Channel (CC1/CC2) pins, each terminated to ground through an independent 5.1 kΩ ± 1% pulldown resistor.
- **Master Power Switch (SW_{102}):** A DPST toggle switch (SW_{102A}, SW_{102B}) placed immediately at the USB-C power entry point. This completely disconnects both the +5 V_{in} supply rail and the primary ground reference (`GNDREF`) when open, preventing quiescent drain and USB host back-powering.

3.2 Active Primary Crowbar Protection (F_{101}, D_{101})

Upstream of the DC-DC converter, an explicit crowbar stage guards against overvoltage transients from off-spec USB adapters:

a. **Series Overcurrent Limiter (F_{101}):** An 1812 surface-mount resettable PTC fuse (**Bourns MF-MSMF110**, I_{hold} = 1.1 A, I_{trip} = 2.2 A, $V_{max} \geq$ 16 V). The 1812 footprint provides low internal resistance (< 0.1 Ω) and high heat dissipation.

b. **Shunt Clamping Diode (D_{101}):** An SMD DO-214AB (SMC) Transient Voltage Suppressor, specifically the **SMCJ6.0A** (V_{RWM} = 6.0 V, V_{BR} = 6.67 V to 7.37 V, 1500 W peak pulse power). A 6.0 V standoff diode clamps immediately above nominal 5 V, driving F_{101} into high-impedance latch-off within milliseconds.

3.3 Isolated DC-DC Converter (U_{101}) & Analog Rail Switch (SW_{101})

- **XP Power JCA0305D03 (U_{101}):** 3 W continuous rating, nominal +5 VDC input (4.5 V to 9.0 V operating window), fully regulated ±15 VDC dual output at ±100 mA.
- **Galvanic Isolation:** 1,500 VDC baseline isolation breaks ground continuity between the computer sound card and floating device-under-test (DUT) circuits.

- **EMI/RFI Containment:** Housed in an integrated nickel-coated copper metal enclosure acting as an intrinsic Faraday shield.
- **Secondary Rail Filtering (C_{103}–C_{106}):** The isolated +15 V and −15 V outputs are filtered by 330 μF 25 V low-ESR radial electrolytics (C_{103}, C_{104}) and local 0.1 μF X7R ceramic bypass caps (C_{105}, C_{106}).
- **Main Power Switch (SW_{101}):** A DPDT switch (SW_{101A}, SW_{101B}) isolates downstream analog circuitry (`+VDC`, `-VDC`) from the DC-DC output, allowing commissioning and power-rail verification without powering active ICs.

3.4 3-LED Diagnostic Monitoring Subsystem

Three dedicated status LEDs provide instantaneous visual verification of system power integrity:

1. **Master Power LED (D_{102}, `M_PWR`):** Positioned on the primary 5 V side after SW_{102}, powered via series resistor R_{101} (470 Ω):

$$I_{\text{LED,pri}} = \frac{5.0\,\text{V} - 2.0\,\text{V}}{470\,\Omega} \approx \mathbf{6.38}\,\text{mA} \tag{1}$$

   Indicates that USB-C power is valid and the master switch is closed.

2. **Positive Analog Rail LED (D_{103}, `PWR_PS`):** Connected between the switched +15 VDC rail and isolated analog ground via R_{102} (3.0 kΩ):

$$I_{\text{LED,pos}} = \frac{15.0\,\text{V} - 2.0\,\text{V}}{3{,}000\,\Omega} \approx \mathbf{4.33}\,\text{mA} \tag{2}$$

3. **Negative Analog Rail LED (D_{104}, `PWR_NEG`):** Connected between isolated analog ground and the −15 VDC rail via R_{103} (3.0 kΩ), sinking ≈ 4.33 mA.

3.5 Secondary TVS Rail Clamping Kickback Protection (D_{105}, D_{106})

During an extreme high-voltage input fault (e.g., 250 V_{RMS} intermittent), fault current shunted through the BAV199 input protection diodes flows directly into the +15 V and −15 V analog power rails.

The Rail Pump-Up Hazard: The XP Power JCA0305D03 is a unidirectional boost-buck flyback converter; its synchronous output rectifiers can only *source* current into the load. They cannot sink reverse current back into the transformer. Injected fault current (≈ 72 mA_{pk}) charges filter capacitors C_{103} and C_{104}, causing the supply rails to rapidly pump upward above ±20 V, destroying the OPA1656 and INA134 op-amps (absolute maximum rating ±18.0 V).

The Solution: Two **Littelfuse SMAJ16A** surface-mount TVS diodes (D_{105}, D_{106}) clamp the rails:

- **Positive Rail Clamp (D_{105}):** Cathode tied to +15 VDC, anode tied to isolated GND. Reverse standoff voltage V_{RWM} = 16.0 V; breakdown V_{BR} = 17.8 V to 19.7 V.
- **Negative Rail Clamp (D_{106}):** Cathode tied to isolated GND, anode tied to −15 VDC.
- **Operation:** Under normal conditions (| ± 15.0 V|), both diodes remain non-conducting with leakage current < 1 μA. During an overvoltage fault, if either rail attempts to rise above 16.5 V, the respective TVS turns on, shunting fault current into ground and clamping the rail below 18.0 V.

3.6 Chassis Ground Isolation Lug Network (J_{102}, R_{104}, C_{107})

To ensure low electromagnetic interference while maintaining complete floating differential safety, an isolated chassis bonding network is provided:

- **Chassis Lug Terminal (J_{102}, `J_chassis`):** Connects mechanically to the extruded aluminum enclosure.
- **High-Voltage Bleeder Resistor (R_{104}):** A 1 MΩ ± 5% 1 W flameproof metal-film resistor shunts electrostatic charge accumulation between the isolated secondary analog ground (AGND) and the metal enclosure.
- **RF Shunt Capacitor (C_{107}):** A 10 nF (0.01 μF) 1 kV ceramic safety capacitor in parallel with R_{104} provides a low-impedance AC return path ($X_C \approx 16\,\Omega$ at 1 MHz) for stray RF pickup and switching EMI.

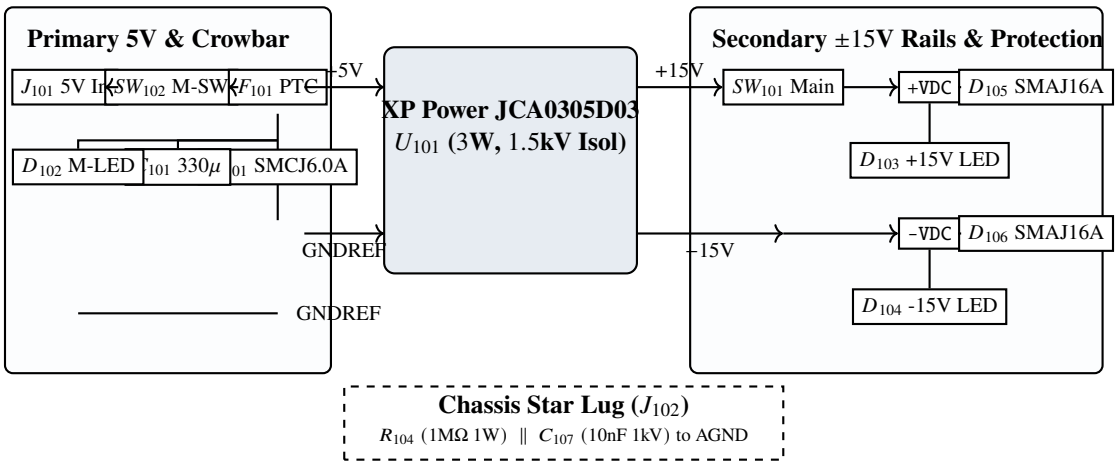


Figure 1: Complete power delivery architecture incorporating SW_{102}, SW_{101}, triple-LED telemetry, SMAJ16A kick-back clamps, and J_{102} chassis lug network.

3.7 Power Budget for Dual-Channel Architecture

Table 4 verifies that current demands across both channels, all three status LEDs, and clamp bias circuits remain comfortably below the 100 mA converter ceiling.

Circuit Block	**Qty**	**Current (±15 V)**	**Operating Condition / Notes**
OPA1656IDR (Dual Op-Amp)	2 ICs	14.0 mA total	High-Z input buffers (Ch 1 & Ch 2)
INA134UA (Diff Amp)	2 ICs	4.8 mA total	Difference amplifiers (Ch 1 & Ch 2)
DRV135UA (Line Driver)	2 ICs	10.4 mA total	Quiescent line driving current
Zener Bias Networks (R_{301}, R_{307})	4 legs	20.6 mA total	Bias current in High-Clamp mode (10.3 mA per channel)
Rail Status LEDs (D_{103}, D_{104})	2 LEDs	8.7 mA total	Continuous visual indicator draw (4.33 mA per rail)
Total Quiescent Draw	—	**58.5** mA	58.5% **of** 100 **mA converter rating**

Table 4: Power consumption budget across dual-channel active components.

4 Analog Front-End (AFE) Signal Conditioning & Protection

4.1 High-Voltage DC Blocking Capacitors (C302, C303, C402, C403)

- **Capacitance Optimization:** Sized at 0.47 μF with a 400 VDC working rating.
- **Cut-off Calculation:** Across the 101.0 kΩ system input impedance, the high-pass -3 dB corner frequency is:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 101{,}000 \times (0.47 \times 10^{-6})} \approx 3.35\,\text{Hz} \tag{3}$$

  At 20 Hz, the magnitude attenuation is below 0.06 dB, and phase shift is negligible.
- **Dielectric Selection:** Metallized Polypropylene (PP) Film (e.g., WIMA MKP4 or Panasonic ECQ-E). High-voltage Class II SMD ceramic capacitors (X7R) are strictly prohibited due to severe Voltage Coefficient of Capacitance (VCC) non-linearities.
- **CAD Footprint Provisioning:** Multi-pitch radial through-hole pads with concentric drill points spaced at 5.0 mm, 7.5 mm, 10.0 mm, and 15.0 mm.

4.2 Active Protection Limiting: The 4.7 kΩ Fusible Resistor

Earlier design revisions considered back-to-back Microchip LND150 depletion MOSFETs to provide current limiting. While attractive for low-frequency current pinching, the LND150 introduces severe limitations in a wideband active probe:

a. **Non-Linear Junction Capacitance:** The LND150 exhibits an input capacitance (C_{iss}) of 15 to 30 pF that varies significantly with instantaneous terminal voltage. Across a 100 kΩ divider impedance, this voltage-dependent capacitance produces harmonic distortion and limits bandwidth below 500 kHz.

b. **Channel Resistance Modulation:** As signal swings exceed several volts, the drain-to-source channel dynamic resistance modulates prior to pinch-off, increasing measured distortion by over 18 dB at 10 kHz.

The Fusible Resistor Solution (R_{312}, R_{313}): The LND150 pair is replaced with a single socketed **Vishay NFR25 4.7 kΩ flameproof fusible resistor**.

4.2.1 Johnson Thermal Noise Analysis

The thermal noise density generated by a resistor is given by:

$$e_n = \sqrt{4k_B T R} \tag{4}$$

At $T = 300$ K with $R = 4.7$ kΩ:

$$e_{n(4.7\text{k})} = \sqrt{4 \times (1.38 \times 10^{-23}) \times 300 \times 4{,}700} \approx \mathbf{8.82}\,\text{nV}/\sqrt{\text{Hz}} \tag{5}$$

Root-sum-squaring this value with the OPA1656 input voltage noise (2.9 nV/$\sqrt{\text{Hz}}$):

$$e_{n(\text{total})} = \sqrt{8.82^2 + 2.9^2} \approx \mathbf{9.28}\,\text{nV}/\sqrt{\text{Hz}} \tag{6}$$

In a standard 20 kHz audio bandwidth, the total integrated input noise is:

$$V_{n,\text{audio}} = 9.28\,\text{nV}/\sqrt{\text{Hz}} \times \sqrt{20{,}000} \approx 1.31\,\mu\text{V}_{\text{RMS}} \tag{7}$$

Referred to a nominal 2.0 V_{RMS} line level, this maintains an input signal-to-noise ratio of 123.6 **dB**, preserving the full dynamic range of the MOTU M4 ADC.

4.2.2 Overvoltage Fault Current & Air-Gap PCB Isolation

During a maximum intermittent 250 V_{RMS} (353 V_{pk}) fault, the peak current shunted into the rail diodes is:

$$I_{\text{fault,pk}} = \frac{353\,\text{V} - 15.6\,\text{V}}{4{,}700\,\Omega} \approx \mathbf{71.8}\,\text{mA} \tag{8}$$

The continuous RMS fault dissipation is:

$$P_{\text{fault}} = \frac{(250\,\text{V})^2}{4{,}700\,\Omega} \approx \mathbf{13.3}\,\text{W} \tag{9}$$

Under this sustained overload, the Vishay NFR25 fuses cleanly within 2 seconds without flame, explosive fragmentation, or carbon tracking.

PCB Creepage Slot: To prevent high-voltage arcing across carbonized FR-4 substrate during a fusing event, a 1.5 **mm** × 6.0 **mm isolation slot** is routed through the PCB (Edge.Cuts) directly beneath the body of R_{312} and R_{313}. Sockets are implemented using Mill-Max gold-plated receptacle pins to allow rapid replacement after a fault.

4.3 Diode Selection: Upgrade from 1N4148 to BAV199

Pete Millett's original design specified standard 1N4148 diodes for rail clamping. These are replaced with **Nexperia BAV199** dual diodes in SOT-23 packages:

- **Reverse Leakage Current (I_R):** A standard 1N4148 leaks 25 to 50 nA at room temperature, increasing to $> 5\,\mu$A at 75°C. When flowing through a high-impedance divider node, this leakage creates significant non-linear DC offsets and measurable odd-order harmonic distortion. The BAV199 guarantees a maximum leakage of 5 **pA at** 25°**C** (< 5 nA at 150°C), completely eliminating leakage-induced distortion.
- **Junction Capacitance (C_j):** The BAV199 maintains $C_j \leq 1.5$ pF under reverse bias, preserving phase margin and high-frequency linearity out to 2.5 MHz.

4.4 Why the Input Attenuator Ladder Must Be Retained

It is tempting to consider bypassing the input resistor divider when using an oscilloscope host, since bench scopes feature their own internal vertical attenuators. This is fundamentally prohibited:

- Probing a high-voltage circuit (such as a power amplifier output or tube plate swinging 150 V to 250 V) without attenuation would drive the OPA1656 inputs far past their ±15 V rails. The BAV199 clamping diodes would conduct continuously, shunting heavy fault current and severely clipping the waveform into a distorted square wave at ±15.6 V.

- The precision stepped attenuator (1×, 10×, 100×, 1000× ranges) scales high-voltage potentials down to the linear operating region of the op-amp buffers, ensuring the buffer and instrumentation amplifier operate within their linear dynamic range.

5 Attenuator Ladder, Switch Selection & Interconnect Wiring

5.1 Why Frequency Compensation Capacitors Are Not Required

Standard oscilloscope passive probes (10×, 100×) use 9 MΩ divider ladders. In those circuits, a 15 pF input capacitance creates an uncompensated pole at ≈ 1.1 kHz, necessitating adjustable trimmer capacitors across every range.

In this active interface, the entire attenuator ladder is scaled down to 101.0 **kΩ total resistance** (90.9 kΩ + 9.09 kΩ + 909 Ω + 102 Ω):

- 1× **Range (Direct Buffer):** No resistive division; buffer input capacitance (≈ 6 pF total) is driven directly by the low-impedance source.
- 10× **Range (Worst-Case Divider Impedance):**

$$R_{\text{th}} = 90.9\,\text{k}\Omega \parallel (9.09\,\text{k}\Omega + 909\,\Omega + 102\,\Omega) \approx \mathbf{9.09}\,\text{k}\mathbf{\Omega} \tag{10}$$

  The total node capacitance (OPA1656 $C_{\text{in}} = 3\,\text{pF}$, BAV199 $C_j = 1.5\,\text{pF}$, stray PCB $C_s \approx 1.5\,\text{pF}$) is ≈ 6.0 pF. The natural uncompensated high-frequency pole is:

$$f_{-3\text{dB}} = \frac{1}{2\pi \times 9{,}090\,\Omega \times 6.0 \times 10^{-12}\,\text{F}} \approx \mathbf{2.92}\,\text{MHz} \tag{11}$$

- 100× **Range:** $R_{\text{th}} \approx 1.0\,\text{k}\Omega \implies f_{-3\text{dB}} \approx \mathbf{26.5}\,\text{MHz}$.
- 1000× **Range:** $R_{\text{th}} \approx 102\,\Omega \implies f_{-3\text{dB}} \approx \mathbf{260}\,\text{MHz}$.

Because the worst-case pole sits comfortably above 2.0 MHz, frequency compensation capacitors across the ladder resistors are unnecessary. This eliminates phase-trim calibration procedures.

5.2 Resistor Power Dissipation & Model Selection

The top ladder resistor ($R_{302} = 90.9\,\text{k}\Omega$) absorbs 90% of the total input voltage.

Operating State	**Input**	**Drop across R_{302}**	**Dissipation**	**Thermal Load Assessment**
120 VAC Mains	120 V_{RMS}	108.0 V_{RMS}	0.128 W	Runs cold (21% of 0.6W rating)
150 V_{RMS} Continuous	150 V_{RMS}	135.0 V_{RMS}	**0.200** W	Safe continuous baseline (33% load)
230 VAC Mains	230 V_{RMS}	207.0 V_{RMS}	0.471 W	Within continuous envelope (78% load)
250 V_{RMS} Intermittent	250 V_{RMS}	225.0 V_{RMS}	**0.557** W	Short-term test rating (93% load, ≤ 5 min)

Table 5: Power dissipation profile across R_{302} under operational conditions.

Selected Model: Vishay Dale CMF55. Axial metal-film series (90.9 kΩ ± 0.1%, 0.6 W, 500 V rating, ±50 ppm/°C). Body dimensions (6.1 mm × 2.3 mm) fit neatly across switch lugs.

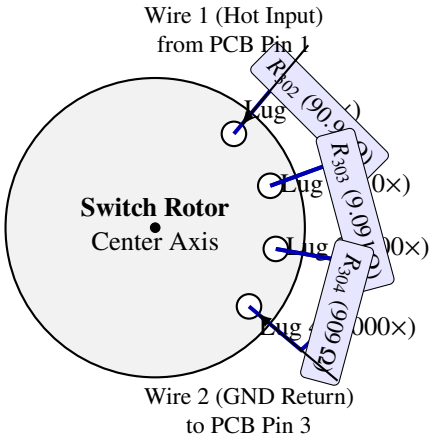
5.3 Rotary Switch Selection: Panel-Mount Models

To eliminate mechanical stress on the PCB and avoid long unshielded trace runs, the attenuator switch mounts to the front panel:

1. **E-Switch KC14A13.001NLS:** Recommended bench value. 1-pole 4-position (SP4T), Break-Before-Make (BBM), solder eyelets, $1,000\,V_{RMS}$ dielectric withstand, 13.0 mm flatted shaft with 9.8 mm threaded bushing. Sits flush behind standard instrument knobs with no shaft trimming required.
2. **C&K A-Series (A10403RNZQ / A10405RNZQ):** 1-pole 4-position (SP4T), 30° indexing, BBM, solder lugs, coin-silver contacts ($< 20\,m\Omega$), $1,000\,V_{RMS}$ dielectric rating. (*Caution: Avoid 'A124' variants, which feature an impractical* 90° *throw per position*).

5.4 Resistor Mounting: Tangential Bending vs. Radial Halo

Switch solder lugs are spaced ≈ 4.0 to 5.7 mm center-to-center. Because the CMF55 resistor body is 6.1 mm long, mounting leads must not be stressed against the body end-caps:



Mounting the resistors in a **radial halo** 6 to 8 mm outside the lug perimeter provides: 1. Complete thermal isolation from the switch body. 2. Direct airflow cooling in open chassis space. 3. Relaxed lead bending clearances with clear PTFE sleeving over exposed copper.

5.5 Why 5-Wire Bundles and Shielded Coax Fail at 2 MHz

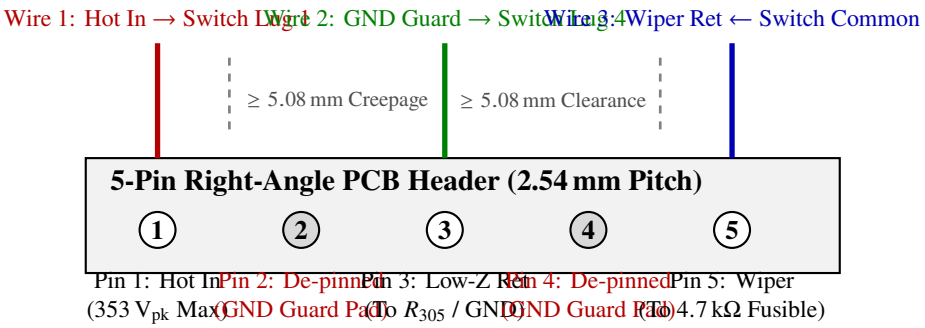
- **Failure of 5 Flying Wires:** Routing 5 unshielded wires from the switch back to the PCB creates mutual stray capacitances (5 to 15 pF) between adjacent leads. High-voltage signals on the 1× lead capacitively bypass the divider ladder and couple directly into the 100× and 1000× taps.
- **Failure of Shielded Coax (RG-178):** Standard RG-178 cable exhibits a capacitance of ≈ 95 pF/m (2.4 pF/inch). A 3-inch run with grounded shield adds ≈ 8 to 10 pF of shunt capacitance to ground. For the 10× tap ($R_{th} = 9.09\,k\Omega$), this rolls off the frequency response at:

$$f_{-3dB} = \frac{1}{2\pi \times 9{,}090\,\Omega \times 16 \times 10^{-12}\,F} \approx \mathbf{1.09}\,\text{MHz} \tag{12}$$

  Grounded coaxial cable degrades bandwidth well below the 2.0 MHz probe requirement.

5.6 The 3-Wire "Guarded Trio" Harness & 5-Pin PCB Header

By placing the divider resistors directly on the switch, **only 3 conductors route to the PCB**. These wires connect to a **5-pin right-angle header** on the board edge:



Header Configuration Rules:

1. **Creepage and Clearance:** Male header pins 2 and 4 are physically extracted from the connector block. This expands the center-to-center clearance between active conductors from 2.54 mm to 5.08 **mm** (0.200″), meeting IPC-2221 requirements for 353 V_{peak} transients.
2. **PCB Guard Trace:** The unpopulated pads for pins 2 and 4 connect directly to the analog ground plane, creating an electrostatic copper shield between the high-voltage trace (Pin 1) and the high-impedance wiper return (Pin 5).
3. **Wire Selection:** Wires are run using **MIL-W-16878/4 Type E stranded PTFE hookup wire** (24 AWG, rated for 600 V_{RMS} continuous, 2,000 V dielectric breakdown). Wire 2 (Low-Z return) is positioned physically between Wire 1 and Wire 3, acting as a floating electrostatic shield without introducing high shunt capacitance to ground.

6 Analog Buffer, Instrumentation Amplifier & CMRR Optimization

6.1 Operational Amplifier Selection: Texas Instruments OPA1656IDR

The dual operational amplifiers (U302, U402) serve as the high-impedance buffer stage. Precision measurements demand that the op-amp introduce negligible loading error and zero DC drift across the attenuator.

Parameter	NE5532 (Legacy Bipolar)	OPA2134 (Legacy JFET)	OPA1656 (Selected CMOS)
Input Bias Current (I_B)	200 nA (typ), 800 nA (max)	5 pA (typ), 100 pA (max)	**10** pA (typ), **25** pA (max)
Voltage Noise Density (e_n)	5.0 nV/$\sqrt{\text{Hz}}$	8.0 nV/$\sqrt{\text{Hz}}$	**2.9** nV/$\sqrt{\text{Hz}}$
Gain-Bandwidth Product (GBW)	10 MHz	8 MHz	**53** MHz
Slew Rate (SR)	9 V/μs	20 V/μs	**24** V/μs
THD+N (1 kHz, 2 kΩ)	0.0005%	0.00008%	**0.000029**% (−131 dB)
Input Offset Voltage Drift	2 μV/°C	2 μV/°C	**0.5** μV/°C
Package / Availability	DIP-8 / SOIC-8	DIP-8 / SOIC-8	SOIC-8 (SMD Surface Mount)

Table 6: Parametric comparison demonstrating the operational advantages of the OPA1656 CMOS amplifier.

DC Offset Analysis: Across the worst-case 100 kΩ input resistance, a traditional bipolar op-amp (NE5532, $I_B \approx 72$ nA) develops an uncalibrated DC offset:

$$V_{\text{offset}} = I_B \times R_{\text{in}} = 72 \times 10^{-9}\,\text{A} \times 100 \times 10^3\,\Omega = 7.2\,\text{mV} \tag{13}$$

In contrast, the OPA1656 generates:

$$V_{\text{offset}} = 10 \times 10^{-12}\,\text{A} \times 100 \times 10^3\,\Omega = 0.001\,\text{mV} \quad (1.0\,\mu\text{V}) \tag{14}$$

This negligible DC offset preserves high DC stability without requiring manual calibration.

6.2 OPA1656 Feedback Optimization for 2 MHz Flatness

The OPA1656 buffer amplifiers operate with local feedback networks ($R_{309}, R_{310} = 7.5\,\text{k}\Omega$):

- Prior designs utilized 10 to 22 pF feedback capacitors, causing bandwidth to roll off at 964 kHz and degrading performance as a 2 MHz scope probe.
- In this revision, feedback capacitors (C_{308}, C_{309}) are set to 4.7 **pF C0G/NP0**.
- The high-frequency feedback pole is shifted to:

$$f_{\text{fb}} = \frac{1}{2\pi \times 7{,}500\,\Omega \times 4.7 \times 10^{-12}\,\text{F}} \approx \mathbf{4.51}\,\text{MHz} \tag{15}$$

  This phase-lead zero stabilizes the buffer loop against board parasitics while maintaining flat frequency response (within −0.3 dB) at 2.0 MHz. Unpopulated 0805 parallel pads (C_{314}, C_{313}) are provided for fine-tuning during testing.

6.3 Elimination of the DC Nulling Network (R_{25})

In the original design, a 100 kΩ cermet potentiometer (R_{25}) was connected to the reference pin (Pin 1) of the INA134 difference amplifier to null DC offsets.

Engineering Justification for Elimination:

1. **Preservation of CMRR:** The Texas Instruments INA134 relies on four internal, on-chip thin-film resistors laser-trimmed to an absolute ratio match better than 0.005%. This precise matching guarantees a Common-Mode Rejection Ratio of > 90 dB across the audio band. Injecting an external DC voltage into Pin 1 introduces source impedance non-linearities and temperature coefficients that unbalance the internal bridge, degrading CMRR by 15 dB to 25 dB.

2. **CMOS Buffer Zero Drift:** The OPA1656 features an input bias current of only 10 pA, reducing offset drift to negligible microvolt levels.

3. **Hardwired Ground Implementation:** Pin 1 of U303 and U403 is hardwired directly to the solid analog ground plane (AGND). Both target audio interfaces and oscilloscopes support software AC coupling, rendering sub-millivolt DC offsets irrelevant.

6.4 Removal of Grounding Switch (SW5) & Monitor Switch (SW3)

Pete Millett's input section included 'SW5' (which shunted the negative input V_{INN} directly to local chassis ground) and 'SW3' (which looped the signal generator back into the input). Both switches have been eliminated:

- **Ground Loop Hazard with Scopes:** Connecting the interface output to the Siglent SDS804X HD references local analog ground to Mains Earth Ground via the scope BNC shield. If 'SW5' were closed while probing a non-zero circuit potential (such as a tube cathode bias network or Class-D bridge output), it would create a **direct short-circuit to Mains Earth**, vaporizing traces or damaging the device under test.

- **Permanent Floating Differential Operation:** By completely removing 'SW5', V_{INN} is routed through DC blocking capacitor C_{303} into the non-inverting input of buffer U302B. The front-end remains permanently floating and differential, providing complete galvanic safety regardless of switch settings.

7 High-Frequency Stability, Layout & Oscillation Prevention

The Texas Instruments OPA1656 is a wide-bandwidth operational amplifier ($f_{\mathrm{unity}} = 53\,\mathrm{MHz}$). At these frequencies, parasitic board capacitances and load cable capacitances can erode amplifier phase margin, resulting in high-frequency ringing or sustained oscillation.

7.1 Active Layout Stability Rules

a. Direct Rail Decoupling: High-frequency surface-mount ceramic bypass capacitors (0.1 μF X7R, 0805) are placed directly against Pin 8 ($V+$) and Pin 4 ($V-$). Power traces route through the capacitor solder pad prior to entering the IC lead frame. A 10 μF tantalum or low-ESR electrolytic capacitor is placed within 10 mm upstream on each rail.

b. Feedback Phase Compensation ($C_{\mathbf{fb}}$): An 0805 capacitor footprint is routed in parallel with the main feedback resistor (R_{fb}). This footprint is populated with a 4.7 pF C0G/NP0 dielectric ceramic capacitor (C_{308}, C_{309}).

c. Series Output Isolation (R_{311}): Driving capacitive coaxial cables directly from low-impedance op-amp outputs degrades phase margin according to $f_{\text{pole}} = \frac{1}{2\pi R_o C_{\text{load}}}$. A dedicated 200 Ω thin-film resistor (R_{311}) isolates the INA134 output from cable capacitance.

7.2 PCB Floorplan & Partitioning Strategy

- **Galvanic Isolation Barrier:** A strict 4.0 mm clearance moat separates the non-isolated USB-C 5 V input domain from the isolated ±15 V analog domain. No copper traces, ground planes, or component pads bridge this barrier except for the footprint of the JCA0305D03 module.
- **Mirrored Symmetry:** Channel 1 and Channel 2 analog signal paths are routed with identical trace lengths and parasitic capacitances, placed symmetrically north and south of a central analog ground bus.
- **Isolated Ground Plane (AGND):** The secondary side of the board utilizes an unbroken, contiguous copper pour on Layer 2 dedicated exclusively to isolated analog ground. This provides low-impedance return paths and minimizes channel-to-channel crosstalk to below −110 dB at 1 kHz.

8 Dual-Level Output Clamping & Universal Driver Subsystem

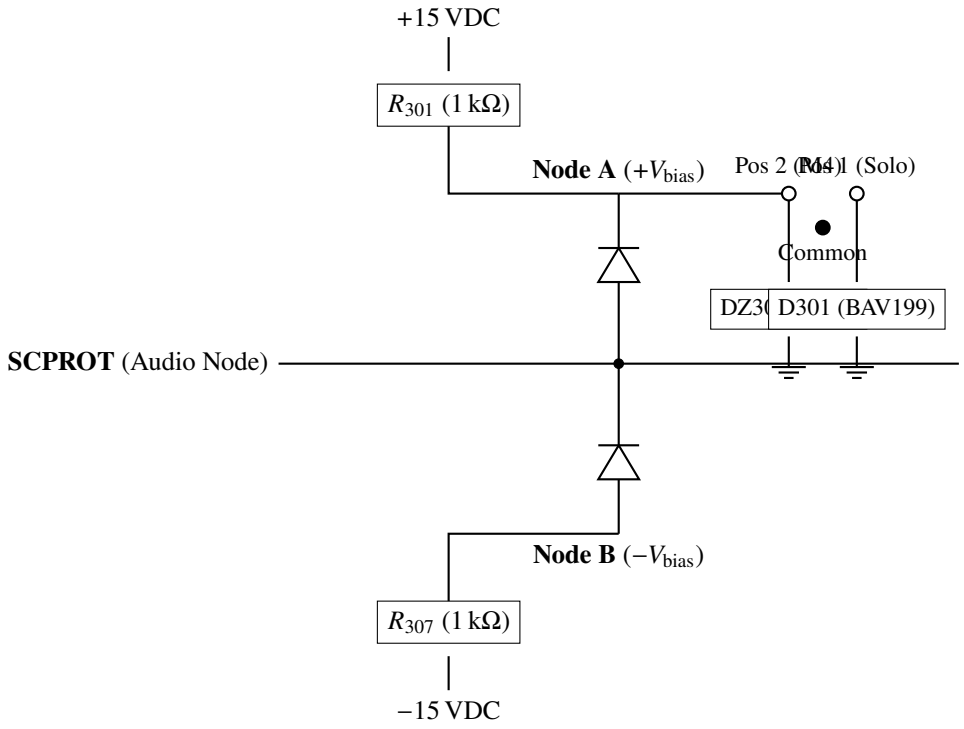
8.1 Why Zener Diodes Cannot Be Placed in the Signal Path

When designing an overvoltage protection clamp for an audio measurement system, clamping diodes cannot simply be connected across the audio line:

- **Junction Capacitance Non-Linearity:** Standard Zener diodes (e.g., 4.7 V Zeners) exhibit massive junction capacitances ($C_j \approx 100$ pF to 400 pF). This capacitance varies non-linearly with applied signal voltage, generating significant harmonic distortion (THD).
- **Soft Reverse Breakdown Knee:** Zener diodes begin conducting microamps of leakage current well below their nominal breakdown rating. On audio peaks above 1 V_{RMS}, this leakage current clips signal peaks roundedly, introducing odd-order harmonics into measurements.

8.2 The Off-Signal Zener Bias Bridge Topology

To achieve high-voltage clamping with zero audio-band distortion, this design maintains ultra-fast, low-capacitance steering diodes (**BAV199**, $C_j < 1.5$ pF) on the audio line while moving the voltage-defining Zeners into the **DC power rail bias legs**:



8.2.1 Clamping Threshold Calculations

1. **Position 1 (M-Audio Solo Mode):**

   - Jumper `JP301/JP302` routes bias nodes to forward-biased diodes ($V_f \approx 0.70\,\text{V}$).
   - Node A is clamped at +0.70 V; Node B is clamped at −0.70 V.
   - Symmetrical signal clipping occurs at:

   $$V_{\text{clamp}} = \pm(V_{\text{bias}} + V_{f(\text{steering})}) = \pm(0.70\,\text{V} + 0.70\,\text{V}) = \pm\mathbf{1.40}\,\text{V}_{\text{peak}} \tag{16}$$

   This limits differential output to $\approx 1.0\,\text{V}_{\text{RMS}}$, protecting the 16-bit M-Audio Solo front-end from digital clipping.

2. **Position 2 (MOTU M4 & Oscilloscope Mode):**

   - The bias nodes connect through **1SMB5917** 4.7 V Zener diodes (DZ_{301}, DZ_{302}).
   - Node A sits at +4.70 V; Node B sits at −4.70 V.
   - Symmetrical signal clipping occurs at:

   $$V_{\text{clamp}} = \pm(4.70\,\text{V} + 0.70\,\text{V}) = \pm\mathbf{5.40}\,\text{V}_{\text{peak}} \quad (\approx \mathbf{3.82}\,\text{V}_{\text{RMS}}) \tag{17}$$

   This takes full advantage of the MOTU M4's +18 dBu ($6.15\,\text{V}_{\text{RMS}}$) line input headroom while protecting the converter.

8.2.2 Bias Resistor Sizing and Noise Suppression

- **Bias Current:** With 4.7 V Zeners active on ±15 V rails:

$$I_{\text{bias}} = \frac{15.0\,\text{V} - 4.70\,\text{V}}{1{,}000\,\Omega} = \mathbf{10.3}\,\text{mA} \tag{18}$$

- **Power Dissipation:** Resistors R_{301} and R_{307} dissipate $(10.3\,\text{mA})^2 \times 1\,\text{k}\Omega = 106\,\text{mW}$ (0.25 W rated). Zener power is $4.7\,\text{V} \times 10.3\,\text{mA} = 48.4\,\text{mW}$, well below the 1.5 W rating of the 1SMB5917.
- **Avalanche Shot Noise Shunting:** Zener diodes generate wideband avalanche noise. Capacitors C_{301} and C_{304} (0.1 μF X7R ceramic) are placed directly in parallel with each Zener to shunt shot noise to AGND.

8.3 Universal Dual Output Stage: DRV135 & Dedicated Scope BNC Tap

Both channels feature two parallel, simultaneously active output connections:

a. Unbalanced BNC Jack ('J303', 'SCPROT'): Directly tapped from the output of the INA134 difference amplifier through series isolation resistor R_{311} (200 Ω thin film). When connected to an oscilloscope with input termination set to 1 MΩ, loading attenuation is negligible:

$$\text{Attenuation} = \frac{1{,}000{,}000\,\Omega}{1{,}000{,}000\,\Omega + 200\,\Omega} = 0.9998 \quad (\mathbf{-0.0017}\,\text{dB}) \tag{19}$$

This feeds directly into the Siglent SDS804X HD oscilloscope or unbalanced line-in soundcards.

b. Balanced $1/4''$ **TRS Jack ('J302'):** Driven by a Texas Instruments **DRV135UA** balanced line driver (SOIC-8). The DRV135 provides an exact laser-trimmed differential gain of 2× (+6 dB), driving long balanced shielded cables into the MOTU M4 rear Line Inputs 3 & 4 with high common-mode ground noise rejection.

9 System Chassis Wiring & Mechanical Integration

To ensure high-frequency stability with the 53 MHz OPA1656 and preserve high CMRR, off-board panel controls must follow strict wiring practices:

Circuit Path	Conductor Type	Routing Specification
Main DC Power (V_{BUS}, GND)	18 AWG Multi-strand Copper	Direct route from USB-C breakout to PCB power terminal; minimizes DC IR drop.
Analog Audio Inputs & Outputs	24 AWG or 26 AWG Shielded Coax	Miniature RG-178 or high-flex shielded audio cable; drain shield tied at PCB star point.
Rotary Attenuator Controls	24 AWG PTFE Discrete Trio	3-wire guarded harness (Wire 2 in center) to 5-pin header; do not twist with coax.
Enclosure Chassis Ground	20 AWG Solid Copper	Direct bond to aluminum chassis star washer lug adjacent to USB-C ground plane.

Table 7: Interconnect and chassis wiring schedule.

9.1 Chassis Construction and RF Shielding

The complete assembly must be enclosed in an extruded aluminum enclosure (e.g., Hammond 1455 series). The aluminum chassis connects to signal ground at a single star point adjacent to the power input. All anodized layers must be cleared with a star washer to ensure low contact resistance between panels and chassis ground. Isolated Teflon front BNC connectors maintain the floating front-end reference.

10 Consolidated Hardware Bill of Materials (BOM)

Designator	Component Function	Selected Part	Package / Specification
U101	Isolated DC-DC Converter	**XP Power JCA0305D03**	3 W, 5 V In, ±15 V Out, 1.5 kV Isolation
U301, U401	Balanced Line Driver (Ch 1, 2)	**DRV135UA**	SOIC-8, Laser-Trimmed $G = 2$ (+6 dB)
U302, U402	Dual High-Z Buffer (Ch 1, 2)	**OPA1656IDR**	SOIC-8, CMOS, 53 MHz, 2.9 nV/$\sqrt{\text{Hz}}$
U303, U403	Difference Amp (Ch 1, 2)	**INA134UA**	SOIC-8, Laser-Trimmed > 90 dB CMRR
D305, D309	Input Rail Steering Clamps	**BAV199**	SOT-23, Dual Diode, $I_R \leq 5$ pA, $C_j \leq 1.5$ pF
D301, D306	Output Steering Clamps	**BAV199**	SOT-23, Dual Diode, $I_R \leq 5$ pA, $C_j \leq 1.5$ pF
DZ301, DZ302	Voltage Bias Zeners	**1SMB5917BT3G**	SMB, 4.7 V, 1.5 W Zener Diode
D101	Primary Crowbar TVS	**SMCJ6.0A**	DO-214AB, 6.0 V Standoff, 1500 W Transient
D105, D106	Secondary Rail Clamping TVS	**SMAJ16A**	DO-214AC, 16.0 V Standoff, Rail Kickback
D102	Master 5V Power LED	**3mm / 0805 Red**	$V_f \approx 2.0$ V, Primary Supply Status
D103, D104	Analog Rail Status LEDs	**3mm / 0805 Green**	$V_f \approx 2.0$ V, ±15 VDC Secondary Status
F101	Resettable Power Fuse	**MF-MSMF110-2**	1812 SMT, 1.1 A Hold, 2.2 A Trip, 16 V

Table 8: Active Silicon, Power & Protection Bill of Materials (Dual-Channel).

Designator	Component Function	Selected Part	Package / Specification
R_{312}, R_{313}	Overvoltage Fault Limiters	**Vishay NFR25**	Axial Fusible, 4.7 kΩ ± 5%, Socketed
R_{302}, R_{402}	Attenuator Top Resistor	**Vishay CMF55**	Axial 0.6 W 500 V, 90.9 kΩ ± 0.1%
R_{303}, R_{403}	Attenuator Middle Resistor	**Vishay CMF55**	Axial 0.6 W 500 V, 9.09 kΩ ± 0.1%
R_{304}, R_{404}	Attenuator Low Resistor	**Vishay CMF55**	Axial 0.6 W 500 V, 909 Ω ± 0.1%
R_{305}, R_{405}	Divider Shunt Resistor	**0.1% Thin Film**	0805 SMD, 102 Ω ± 0.1%
R_{309}, R_{310}	Buffer Feedback Resistors	**0.1% Thin Film**	0805 SMD, 7.5 kΩ ± 0.1%
R_{308}	Buffer Gain Resistor	**0.1% Thin Film**	0805 SMD, 10.0 kΩ ± 0.1%
R_{311}	BNC Series Isolation Resistor	**1% Thin Film**	0805 SMD, 200 Ω ± 1%
R_{301}, R_{307}	Clamp Pre-Bias Resistors	**1% Thin Film**	0805 SMD, 1.0 kΩ ± 1%
R_{101}	Master LED Dropping Resistor	**1% Thin Film**	0805 SMD, 470 Ω ± 1% (0.125 W)
R_{102}, R_{103}	Rail LED Dropping Resistors	**1% Thin Film**	0805 SMD, 3.0 kΩ ± 1% (0.125 W)
R_{104}	Chassis Bleeder Resistor	**Flameproof Metal Film**	Axial 1 W, 1.0 MΩ ± 5% High-Voltage
C_{302}, C_{303}	DC Blocking Capacitors	**0.47 μF 400 V**	WIMA MKP4 Metallized Polypropylene
C_{308}, C_{309}	2 MHz Feedback Capacitors	**4.7 pF C0G/NP0**	0805 SMD, 50 V Ceramic (±0.25 pF)
C_{313}, C_{314}	Feedback Tuning Pads	**DNP**	0805 Unpopulated Footprints
C_{301}, C_{304}	Zener Noise Shunt Capacitors	**0.1 μF 50 V**	0805 X7R Ceramic Bypass
C_{102}, C_{105}, C_{106}	DC-DC Switching Snubbers	**0.1 μF 50 V**	0805 X7R Ceramic Bypass
C101	Primary Bulk Input Decoupling	**330 μF 25 V**	Low-ESR Radial / Aluminum Electrolytic
C103, C104	Secondary Output Rail Filters	**330 μF 25 V**	Low-ESR Radial / Aluminum Electrolytic
C107	Chassis Ground Safety Cap	**10 nF 1 kV**	Radial Disc / SMD 1812 Ceramic, 1000 V
SW102	Master Primary Power Switch	**DPST / DPDT Toggle**	Panel Mount, 5V/GND disconnect
SW101	Secondary Dual-Rail Switch	**DPDT Sub-Mini**	Panel Mount, ±15 VDC disconnect
SW301, SW401	Panel Attenuator Switches	**KC14A13.001NLS**	E-Switch 1P4T BBM, Solder Lugs, 13 mm Shaft
JP301, JP302	Clamping Mode Selectors	**3-Pin Header**	2.54 mm Shunt Jumper or Sub-Mini Toggle
J101	DC Power Input	**USB-C 6-Pin**	Surface Mount Power-Only with 5.1 kΩ pulldowns
J102	Chassis Ground Lug	**Turret / Eyelet Lug**	M3 / M4 Chassis Star-Washer Terminal
J301, J401	Front Signal Inputs	**Isolated BNC**	Amphenol Bayonet, Isolated Chassis Bushing
J302, J402	Balanced Audio Outputs	**1/4" TRS Jack**	Neutrik / Rean NYS216 Isolated Jack
J303, J403	Wideband Scope Outputs	**Chassis BNC**	Isolated BNC Tap through R_{311} (200 Ω)

Table 9: Passive Components & Electro-Mechanical Hardware Bill of Materials.

11 Bench Commissioning & Verification Workflow

Once fabricated, the dual-channel interface should be validated following this step-by-step procedure:

1. **Cold Resistance & Isolation Verification:** With power disconnected and SW_{102} open, verify $> 100\,\text{M}\Omega$ insulation resistance between the USB-C shield and the analog BNC shells using a high-voltage insulation tester or high-range DMM. Ensure Pin 1 of both INA134 ICs measures $< 0.1\,\Omega$ to isolated AGND. Verify that J_{102} measures $1.0\,\text{M}\Omega$ to isolated AGND.

2. **Power Rail Check:** Close Master Switch SW_{102} while keeping Secondary Switch SW_{101} open. Verify LED D_{102} illuminates. Check test points TP101 and TP102 relative to TP103: rails must measure precisely between $\pm 14.85\,\text{V}$ and $\pm 15.15\,\text{V}$ with under $5\,\text{mV}_{\text{p-p}}$ switching ripple on the oscilloscope. Close SW_{101} and verify LEDs D_{103} and D_{104} illuminate symmetrically.

3. **Output Clamping Calibration:**

   - Set JP301/JP302 to Low (Solo Mode). Inject a 1 kHz sine wave into the front BNC and increase generator level. Observe on the SDS804X HD that the waveform begins symmetrical hard clipping at precisely $\pm 1.40\,\text{V}_{\text{peak}} \pm 0.05\,\text{V}$.
   - Set JP301/JP302 to High (M4/Scope Mode). Verify that the clipping threshold shifts upward to $\pm 5.40\,\text{V}_{\text{peak}} \pm 0.15\,\text{V}$ without distortion on signals below $3.5\,\text{V}_{\text{RMS}}$.

4. **Host Connection Protocols:**

   - *M-Audio Solo:* Connect rear BNC out to Solo Input 2 set to 'Line', with clamp switch set to **Low**.
   - *MOTU M4:* Connect rear $1/4''$ TRS jacks to rear Line Inputs 3 & 4 via TRS-to-TRS balanced cables, with clamp switch set to **High**.
   - *Siglent SDS804X HD:* Connect rear BNC jacks to Scope Ch 1 and Ch 2 via $50\,\Omega$ coaxial cables, with clamp switch set to **High** and scope vertical channel set to $1\,\text{M}\Omega$ input impedance.

12 Summary & Next Steps

This updated specification establishes a low-noise, dual-channel active differential probe and conditioning front-end serving the M-Audio Solo, MOTU M4, and SDS804X HD.

The immediate follow-on engineering tasks are:

1. Generate the complete schematic netlist in KiCad including all unified power, protection, attenuator, and buffer sheets.

2. Finalize the mechanical drilling template for the Hammond extruded enclosure front/rear panels.

3. Execute PCB routing following the strict 2-layer star-grounding and high-frequency decoupling layout guidelines outlined in Section 7.